

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (ME/CL/EE)

November-December 2013

IT 101 – Fundamentals of Computer Programming

Date: 30.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 (a) Do as directed.

1. Convert the binary number 11010111101110 to its equivalent Octal and Hexadecimal numbers. [02]
2. Draw the flow chart to find minimum among the three integers. [03]

(b) What is the correct output of the following code? [06]

(1) main() { int a=3, b=4, c; c = a++; b=c++; printf("%d %d", a,c); }	(2) main() { char c='D', p, q; p=c+2; q=c-2; printf("%c %c", p, q); }	(3) main() { int cnt=0, S=4; while(cnt < S) { printf("%d",cnt); cnt++; }}
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Q - 2 (a) Explain the basic structure of a C program. [05]

(b) Explain the advantages of use user-defined functions in a program. [05]

(c) Write a program to print multiplication of two floating numbers. [02]

OR

Q - 2 (a) Explain the entry-controlled loop and exit-controlled loop in detail with example. [05]

(b) Explain *Call by value* and *Call by address* with example. [05]

(c) Write a program that calculates the average of three integer numbers. [02]

Q - 3 1. Write a C program that evaluate the following series and print the answer. [12]

$$1^2 - 2^2 + 3^2 - 4^2 + \dots \pm n^2$$

2. Write a C program that takes a string from keyboard and find the occurrence of entered character by the user. (For example: if the entered string is *indian*, and user asked to check the occurrence of *i* then program should print 2.)

OR

Q - 3 1. Write a C program that prints all the numbers divisible by 4 between N1 to N2, [12]
where N1 and N2 are user dependent.

2. Write a C program that evaluate the expression: $nCr = \frac{n!}{r!(n-r)!}$

To calculate factorial of a number define *fact()* function that take an integer an argument and returns its factorial.

SECTION – II

Q - 4 (a) Mention whether the following statements are True or False. [05]

1. There is only one ternary operator in C.
2. Negative integer cannot be used with modulo operator.
3. Global variables must be defined inside the body of the function.
4. Pointer is a variable that is used to store the address of a variable.
5. #define is used to define symbolic variable.

Q - 4 (b) Do as directed. [06]

1. Classify valid and invalid variable name in C.
(i) ANSI (ii) avg_Max (iii) Ra.1
(iv) modulo (v) input_scanf (vi) 13saath
2. Evaluate the following expression according to C. Show each step.

$$4 + 2 * 3 - 5 / 2 + 1 * 2$$

Q - 5 (a) 1. Write a C program that takes number from 1 to 7. It prints day accordingly. For [10]
example, if user enters 1 then it displays Monday, if 5 then Friday and so on. Use switch statement for this.

2. Write a C program that inputs 8 floating numbers of an array and print the addition of them.

(b) How to input a sentence (words with space) in C language? [02]

OR

Q - 5 (a) 1. Write a C program that checks whether the entered number is palindrome or not. [10]
2. Define structure named Student with members: int id, mark1, mark2. Write a C program that enters data for 5 students using structure and print individual total marks of each student.

(b) What is infinite loop? Write an infinite loop using *for* statement. [02]

Q - 6 (a) Differentiate with example: Structure vs Union [04]

(b) Answer the following questions. [08]

1. What is unary and binary operator? Given the example.
2. What is pointer? What is the address of a variable?
3. Explain *strlen()* and *strcpy()* functions.
4. Define structure that is appropriate to store a date. Also declare a variable today.

OR

Q - 6 (a) Differentiate with example: Local variable vs Global variable [04]

(b) Answer the following questions. [08]

1. Explain the working of *goto* statement.
2. Explain the conditional operator.
3. Explain recursion with example.
4. Explain any two functions of *ctype.h* file.
